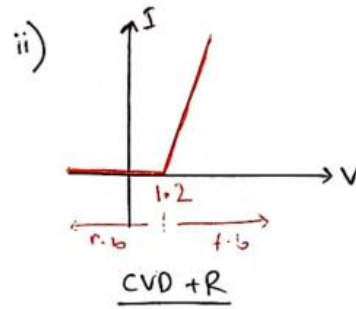
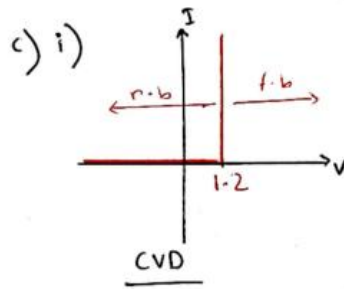


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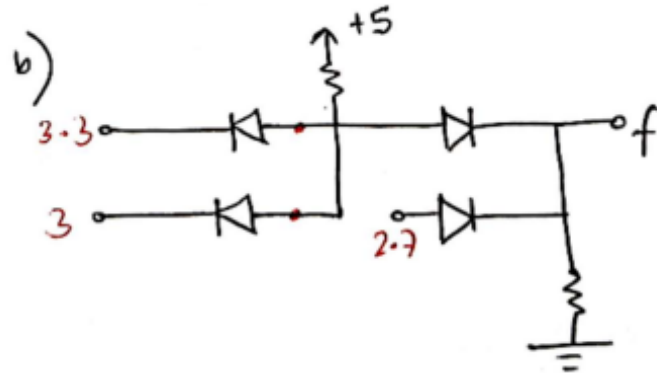
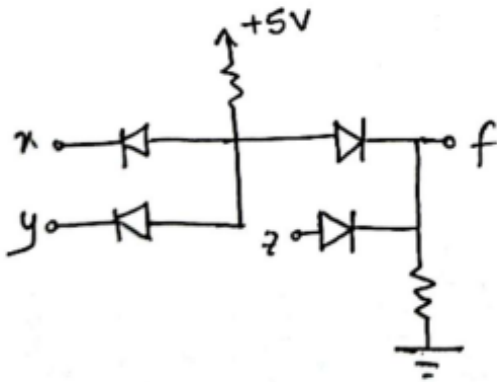
CSE 251 Mid Solution (Set A)

1) a) $R_i = \infty$
 $R_o = 0$
 $A = \infty$



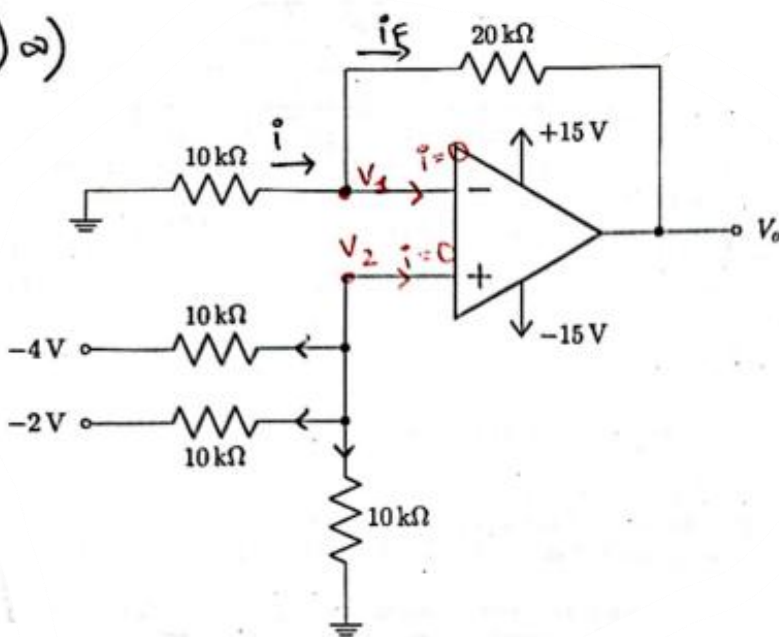
b) $V^+ \approx V^-$ only under closed-loop configuration with negative feedback. This concept is called virtual ground.

2) a) $f = xy + z$



o/p of $x \cdot y = \min(3.3 + 0.7, 3 + 0.7) = 3.7V$
o/p of $x \cdot y + z = \max(3.7 - 0.7, 2.7 - 0.7) = 3.0V$

3) a)



KCL at V_2 ,

$$\frac{V_2 - (-4)}{10} + \frac{V_2 - (-2)}{10} + \frac{V_2}{10} = 0$$

$$\Rightarrow V_2 = -2V$$

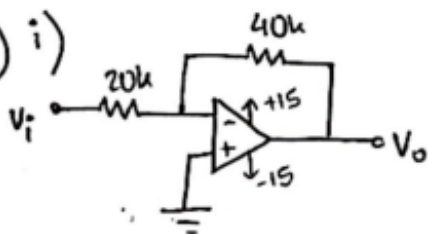
KCL at V_1 ,

$$i = i_F$$

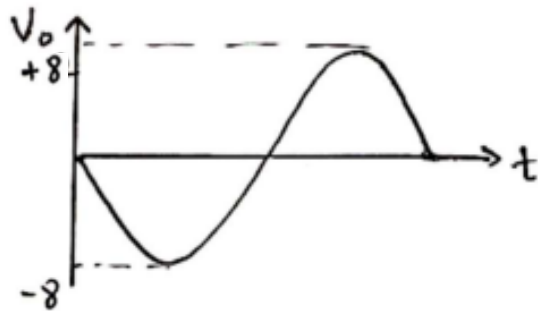
$$\Rightarrow \frac{0 - V_1}{10} = \frac{V_1 - V_o}{20} \quad [V_1 = V_2 = -2]$$

$$\Rightarrow V_o = -6V$$

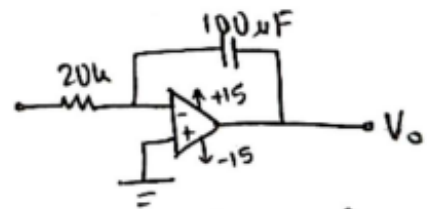
3) b) i)



$$V_o = -\frac{40}{20} V_i = -8 \sin(t)$$

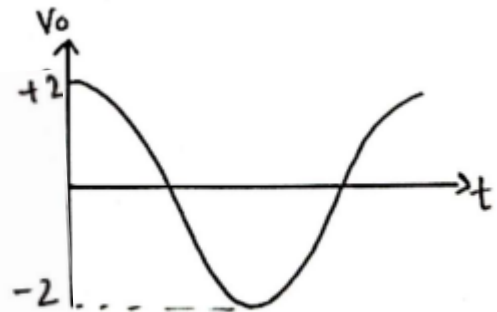


ii)

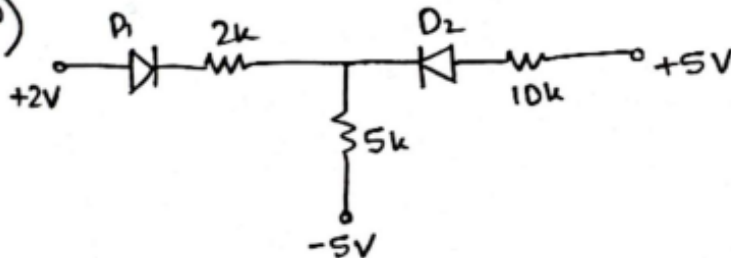


$$V_o = -\frac{1}{20k \times 100\mu F} \int 4 \sin t \, dt$$

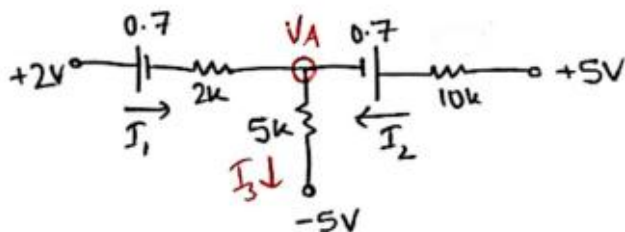
$$= -(-2 \cos t) = +2 \cos t$$



4) a)



b) Assume D_1 ON, D_2 ON



KCL at V_A ,

$$I_1 + I_2 = I_3$$

$$\Rightarrow \frac{2 - V_A - 0.7}{2} + \frac{5 - V_A - 0.7}{10} = \frac{V_A + 5}{5}$$

$$\therefore V_A = 0.1 \text{ V}$$

$$I_1 = \frac{2 - V_A - 0.7}{2k} = 0.6 \text{ mA} > 0$$

$$I_2 = \frac{5 - V_A - 0.7}{10k} = 0.42 \text{ mA} > 0$$

$\therefore$ Assumptions considered are

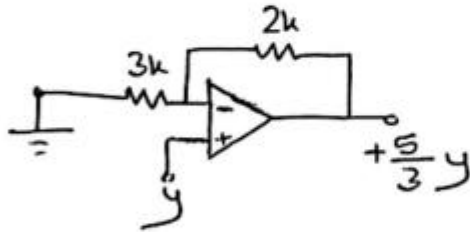
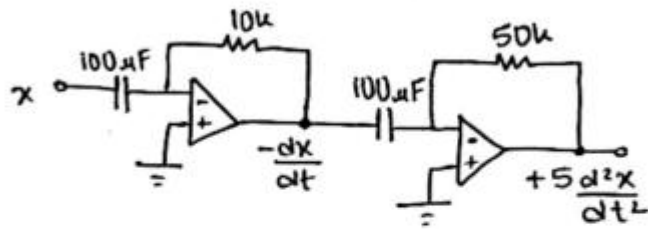
CORRECT.

$$P_{D1} = I_1 \times V_{D1} = 0.6 \times 0.7 = 0.42 \text{ mW}$$

$$P_{D2} = I_2 \times V_{D2} = 0.42 \times 0.7 = 0.294 \text{ mW}$$

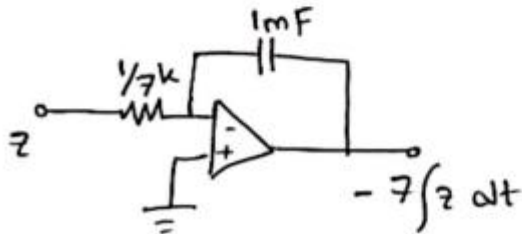
$$5) \infty) f = -5 \frac{d^2}{dt^2} x - \frac{5}{3} y + 7 \int z dt$$

$$= -\left(5 \frac{d^2 x}{dt^2}\right) - \left(\frac{5}{3} y\right) - \left(-7 \int z dt\right)$$



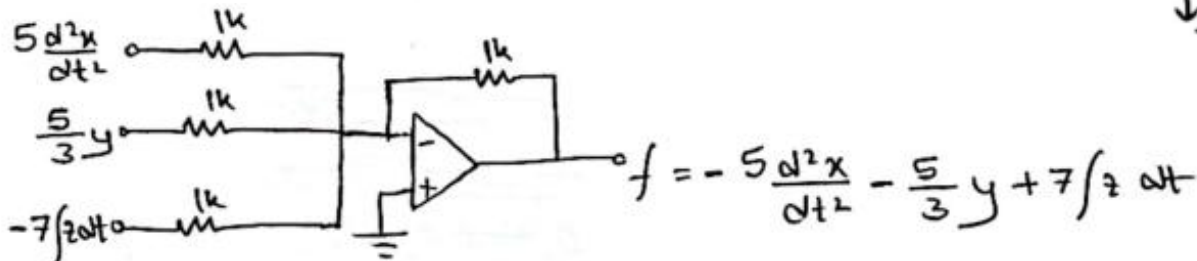
$$1 + \frac{R_f}{R_i} = \frac{5}{3}$$

$$\therefore \frac{R_f}{R_i} = \frac{2}{3}$$

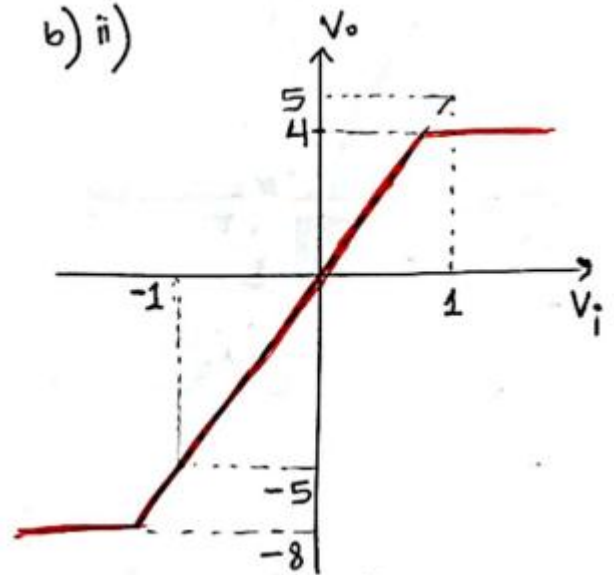


$$+ \frac{1}{RC} = +7$$

choose any R, C.



b) ii)

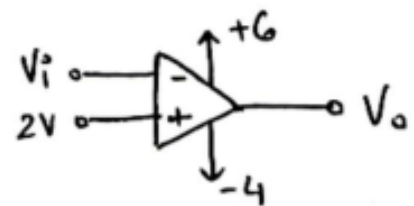


$$c) V_{ref} = 2$$

$$V_i < 2, V_o = +6V \quad \left. \begin{array}{l} V_o = +V_{sat}, V_2 > V_1 \\ V_o = -V_{sat}, V_2 < V_1 \end{array} \right\}$$

$$V_i > 2, V_o = -4V$$

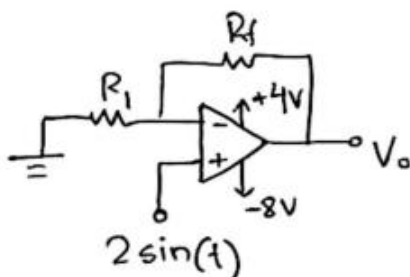
$$\therefore V_2 = 2V, V_1 = V_i$$



$$5) b) i) \text{ gain} = \frac{o/p}{i/p} = \frac{10}{2} = 5$$

max of observed o/p = +4V
min of " " = -8V } saturation voltages

Amplifier type - non-inverting (0° phase change)



$$1 + \frac{R_f}{R_i} = 5$$

$$\frac{R_f}{R_i} = 4$$

$$\therefore \text{Let } R_i = 1k, R_f = 4k$$